



Lab 2: Kirchhoff's Voltage Law (KVL) and Voltage Divider Rule (VDR) using Series Circuit

Objectives

- Learn how to connect a series circuit on a breadboard.
- Validate the Voltage Divider Rule.
- Verify Kirchhoff's Voltage Law.

Theory:

Voltage Division Rule: Consider the circuit given below where two resistors are in series, since the same current i flows in both of them.

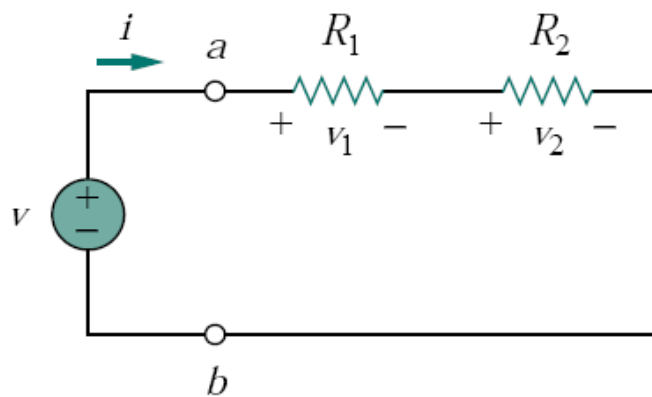


Fig 1: A single-loop circuit with two resistors in series

Applying Ohm's law to each of the resistors, we obtain,

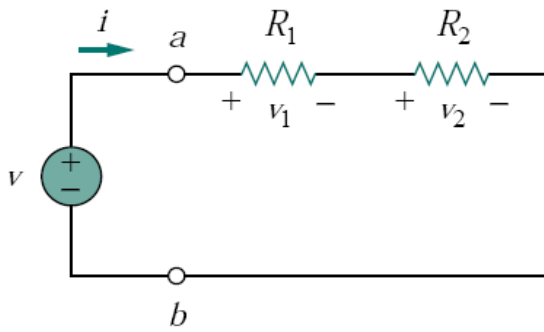
$$v_1 = iR_1, \quad v_2 = iR_2$$

The Principle of voltage division is that the source voltage v is divided among the resistors in direct proportion to their resistances; the larger the resistance, the larger the voltage drop. The circuit in Fig.1 is called a **voltage divider**.

$$v_n = \frac{R_n}{R_{eq}} v = \frac{R_n}{R_1 + R_2 + \dots + R_N} v$$

Applying the voltage division rule we can now determine the voltage across each resistor.

For two resistors in series:

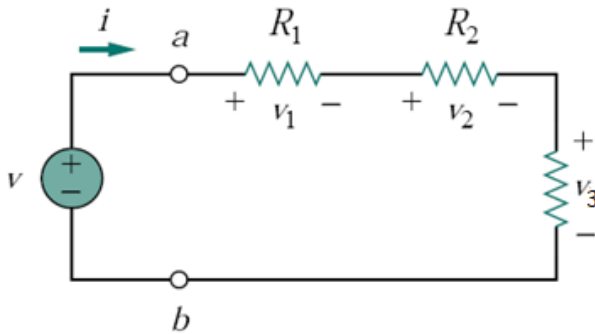


In this circuit:

$$v_1 = \frac{R_1}{R_1 + R_2} v$$

$$v_2 = \frac{R_2}{R_1 + R_2} v$$

For three resistors in series:



In this circuit:

$$R_{eq} = R_1 + R_2 + R_3$$

$$v_1 = \frac{R_1}{R_{eq}} v$$

$$v_2 = \frac{R_2}{R_{eq}} v$$

$$v_3 = \frac{R_3}{R_{eq}} v$$

Kirchhoff's voltage law (KVL): Kirchhoff's voltage law states that the algebraic sum of all voltages around a closed path (or loop) is zero.

Expressed mathematically, KVL states that,

$$\sum_{m=1}^M v_m = 0$$

Where "m" is the number of voltages in the loop and v_m is the m^{th} voltage.

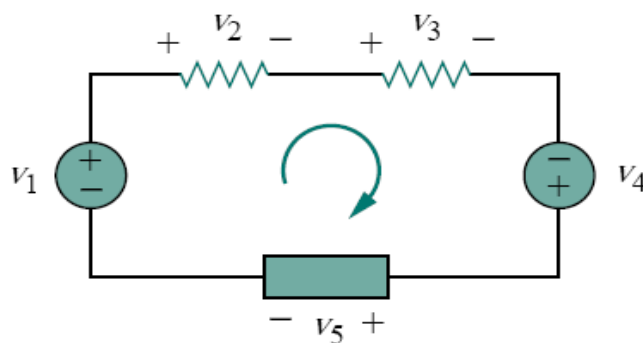


Fig 2: A single-loop circuit illustrating KVL

Applying KVL, we can find that,

$$-v_1 + v_2 + v_3 - v_4 + v_5 = 0$$

Rearranging terms gives,

$$v_2 + v_3 + v_5 = v_1 + v_4$$

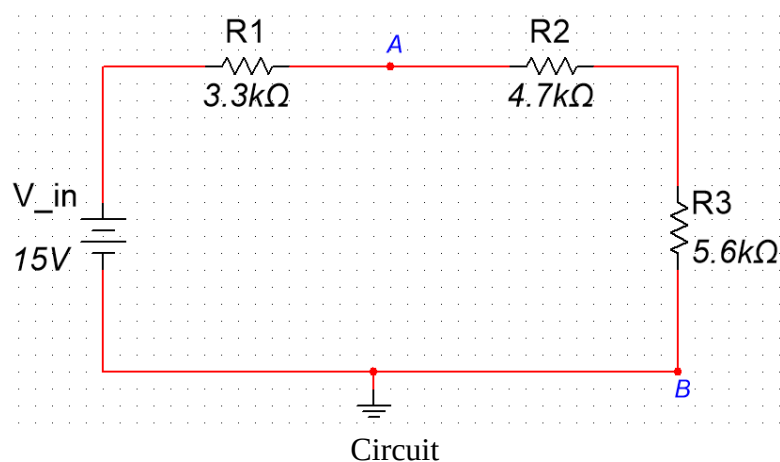
which may be interpreted as

Sum of voltage drops = Sum of voltage rises

List of Components:

Component Name	Value (or Model Number)	Quantity
Breadboard		1
DC Power Supply (Variable)		1
Resistors	3.3 k Ω	1
	4.7 k Ω	1
	5.6 k Ω	1
Connecting Wires		10
Digital Multimeter (DMM)		1

Circuit Diagram:



Procedure:

1. Measure the resistances of the resistors using the DMM and fill in the required column in Table 1.
2. Calculate the percentage error of the resistance values.

$$\text{Percentage of Error} = \frac{|(\text{Practical value} - \text{Theoretical value})|}{\text{Theoretical value}} \times 100\%$$

3. Build the circuit.
4. Using the DMM, find the potential differences across the source V_S and resistors R_1 , R_2 and R_3 . Record the readings in Table 2.
5. Fill in Table 3.
6. Measure V_{ab} . Calculate V_{ab} using voltage division rule. Note down values in Table 4.
7. Now, disconnect the voltage source from the circuit and measure the total load resistance, R_{eq} of the circuit using DMM. Note down values in Table 4.

Data Collection:

Lab 2:

Group No. _____

Instructor's Signature _____

Table 1:

Theoretical Value	Practical Value using DMM	% of Error
3.3 k Ω		
4.7 k Ω		
5.6 k Ω		
$R_T = R_1 + R_2 + R_3 = 13.6 \text{ k}\Omega$		

Table 2:

	Experimental readings	Theoretical values	% of Error
V_{in}			
V_{R1}			
V_{R2}			
V_{R3}			

Table 3: use data from table 2

Potential rise V_{in}		Are the voltage rises and drops equal?
Sum of potential drops $V_{R1} + V_{R2} + V_{R3}$		☐ Equal ☐ Almost Equal ☐ Not Equal

Table 4:

	Experimental readings	Theoretical values	% of Error
V_{ab}			

Questions

1. State the voltage division rule (VDR).
2. State the Kirchhoff's voltage law (KVL).
3. Showing all the steps, calculate R_{eq} . Compare with the experimental value.
4. Showing all steps, calculate the theoretical values in Table 2. Compare theoretical values to your experimental values and explain whether your circuit follows KVL or not.
5. Showing all the calculations, theoretically calculate V_{ab} . Compare with the experimental value and verify the voltage division rule at the terminal a-b.

MultiSim Simulation

- Simulate the KVL circuit you did in lab class. Fill the data tables. Verify KVL.